ALBIN JOJO

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Objective

Electronics and Computer Engineering student (3rd Year) with hands-on project experience in embedded systems, IoT, and edge AI deployment on resource-constrained hardware. Seeking an embedded systems internship to build deeper expertise in firmware development and hardware–software co-design.

Education

Saintgits College of Engineering, Kottayam 2023 – Present
B.Tech in Electronics and Computer Engineering CGPA 7.13
Affiliated to APJ Abdul Kalam Technological University (KTU)

Experience

Embedded Systems Intern | Keltron (Kerala State Electronics Development Corporation) 2024 (2 weeks)

- Completed structured industry training on firmware development for ESP32 and Raspberry Pi at a state-owned electronics PSU.
- Gained practical exposure to I2C, UART, and SPI protocol implementation and industrial sensor-actuator interfacing.

Technical Skills

Microcontrollers & Hardware: Arduino, ESP32, ESP8266, Raspberry Pi
Programming Languages: C, Python, Verilog
Operating Systems: Linux, Raspberry Pi OS
AI / ML: PyTorch, YOLOv8, MobileNetV3, ONNX Runtime, OpenCV
Protocols: I2C, UART, SPI, MQTT, SPIFFS
Tools: Git, PlatformIO, Icarus Verilog, GTKWave, KiCad

Projects

Tomato Leaf Disease Edge Classifier 2026
YOLOv8s · MobileNetV3-Large · ONNX Runtime · Raspberry Pi 4B · Edge AI github.com/albinnnnn/leaf-disease-detector

- Built a two-stage inference pipeline (YOLOv8s leaf detector + MobileNetV3-Large 10-class classifier) exported to FP32 ONNX for fully offline deployment on Raspberry Pi 4B.
- Achieved approximately 600 ms per-frame inference with no cloud dependency; weighted F1 score of 0.93 across 10 disease classes on a 14,500-image test set.
- Implemented modular inference system with daemon camera thread, interrupt-driven GPIO via gpiozero, and direct framebuffer rendering on a 3.5-inch TFT display.

FSM-Based UART Transceiver with 16x Oversampling 2026
Verilog · FSM · Digital Design · Icarus Verilog · GTKWave github.com/albinnnnn/verilog-uart-fsm

- Designed a complete FSM-based UART TX/RX module in Verilog, including a parameterised baud rate generator with 16x oversampling for robust mid-bit RX sampling.
- Verified start/stop bit framing, serial-to-parallel conversion, and bit-sampling alignment via loopback testbench simulation; waveforms validated in GTKWave at 9600 baud.

Distributed RFID Attendance System 2025
ESP32 · MQTT · SPIFFS · Google Sheets API · IoT github.com/albinnnnn/edge-based-rfid-attendance-system

- Architected a 3-node IoT network over MQTT — 2 dedicated RFID scanner nodes and 1 aggregator node — for real-time attendance synchronisation and centralised logging via Google Sheets API.
- Implemented SPIFFS-backed offline queuing on each node, ensuring zero data loss during Wi-Fi outages with automatic sync on reconnect.

Smart Reading Glasses — Assistive OCR Device 2024
Raspberry Pi Zero W · Tesseract OCR · Bluetooth · Text-to-Speech · Linux github.com/albinnnnn/ai-assisted-smart-glasses

- Built a wearable offline OCR and Text-to-Speech device on Raspberry Pi Zero W; adaptive image thresholding improved Tesseract recognition accuracy over baseline.
- Optimised memory-constrained pipeline to achieve 2–3 s end-to-end latency per capture with no cloud dependency.

Certifications and Training

Winter Technical Training: Drone Technology and Remote Sensing Feb 2026
India Space Lab — Sensor fusion, telemetry communication, and UAV systems integration.

Linux and Shell Scripting: Professional Certificate Oct 2025
IBM / Coursera — Bash scripting and Linux system administration.

Vega Processor Workshop: SoC and RISC-V Architecture May 2024
CDAC Trivandrum — RISC-V instruction set architecture and SoC design fundamentals.